» tough and resilient as other synthetics but much thinner and lighter. The lightness and durability of microfibers was first used in suits for astronauts, but is now found in sportswear. This trend for "high end" technology to filter down into everyday life is common. The polymer Teflon—discovered accidentally during a refrigeration experiment—is used to insulate coat bullets, but also forms the surface of a nonstick frying pan.

Technology in everyday life

Breakthrough inventions of the 20th and early 21st centuries have entered everyday usage as a result of mass production techniques. The invention of household machines has led to an increase in leisure time in developed countries and the very concept of progress has become synonymous with the ready availability of new technology. The development of the microchip has had a huge impact on the way we communicate since being pioneered by Jack Kilby and Robert Noyce in the 1950s (see p.479). Miniaturized technology combined with the advent of the Internet has provided access to global instant communication.

> "The only way to discover the **limits of the possible** is to venture a little past them into the **impossible.**"
>
> ARTHUR C. CLARKE, WRITER, 1962

BAIRD MECHANICAL TELEVISION, MANUFACTURED 1926

NYLON

35MM CAMERA, MANUFACTURED 1929

ELECTRON MICROSCOPE

INVENTIONS

New technologies invented during the last 100 years have changed the world for ever.

1920

1920 The **electric food mixer** is produced by the Hobart Corporation, providing a smaller version of industrial mixers for use in the home.

1925 John Logie Baird creates the **first mechanical television**, successfully transmitting pictures in his London attic.

1927 The first **flash bulb** is invented by Paul Vierkotter, allowing photographers to take pictures without natural light.

1930

1930 Frank Whittle submits his plans for a **turbojet aircraft engine** to the RAF.

1931 Max Knott and Ernst Ruska invent the first **electron microscope**. It uses electrons to create a magnified image.

1934 Percy Shaw invents the **cat's-eye**, a safety device used in road construction.

1934 Joseph Begun invents the first **tape recorder**, capable of storing and playing back sound.

1935 Wallace Carrothers and DuPont Labs invent **nylon**, the first truly **manmade fiber**.

1940

1940s Carbon fiber is developed in a British laboratory. It is light, durable, and extremely versatile. Modern applications include engineering products and motor sports, sailing, and cycling equipment.

1945 The atomic bomb is invented by scientists working on the US Manhattan Project.

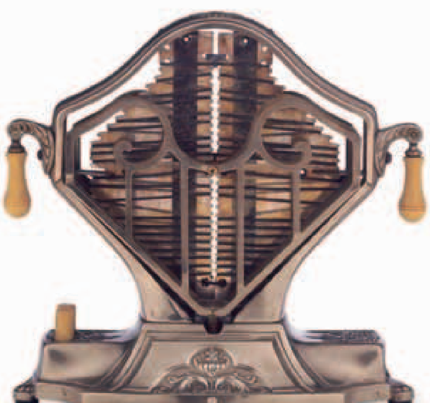
1948 The **vinyl LP** is introduced. Records used to be made from glass or zinc. Vinyl improved durability and sound quality.

1950

1953 The **portable transistor radio** is invented by Texas Instruments.

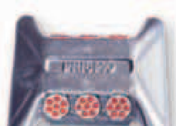
1954 Chaplin, Fuller, and Parson invent the first **solar panels** using photovoltaic cells to convert the Sun's energy into electrical energy.

1959 Jack Kilby and Robert Noyce invent the **microchip**.



ELECTRIC TOASTER, MANUFACTURED 1920

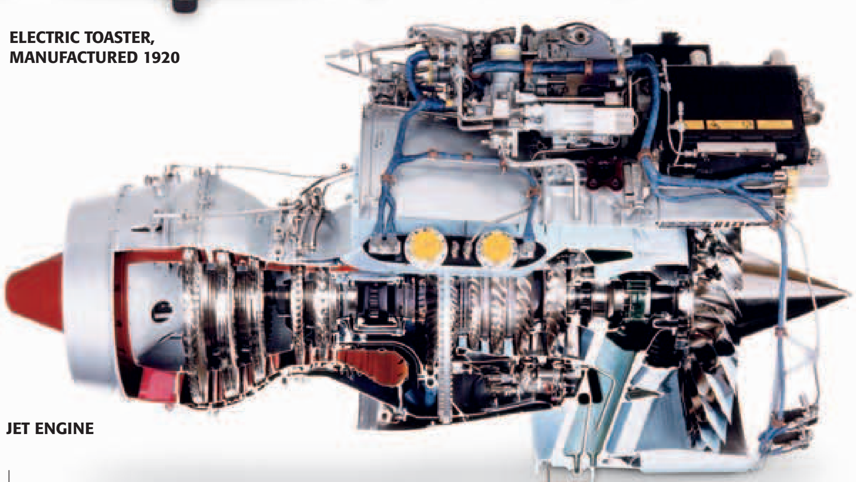
VINYL RECORD



CAT'S-EYE

CARBON FIBER BIKE

AUDIO TAPE PLAYER, MANUFACTURED 1950



JET ENGINE

ENGINEER (1907–96)

FRANK WHITTLE

Frank Whittle joined the British Royal Air Force as an aircraft apprentice, at the age of 17. During his training he wrote a thesis proposing that piston aircraft engines be replaced by turbines. In 1936, Whittle set up the company Power Jets Ltd. and developed these ideas, creating the "Gloster Whittle," the first jet engine aircraft to fly in the UK. He worked with engineering manufacturers, such as Rolls Royce, and then became a research professor at the US Naval Academy.